

UTAH PRISON PROJECTIONS PROJECT: PROCESS & METHODOLOGY - 2025 UPDATE



Overview: This document outlines the process and statistical methodology used to forecast Utah's prison population. The framework is designed to be dynamic and adaptable, enabling the integration of new data and adjustment to evolving trends.

Introduction

The Utah Prison Projection Project (UPPP) is a collaborative effort between the Utah Department of Corrections (UDC) and the Utah Department of Criminal Justice (DOCJ). This document provides an overview of the process and statistical approach employed to forecast Utah's prison population. Future projection efforts will be designed to adapt to changes in policies and legislation, and to allow for deeper analysis of key sub-populations. The full UPPP project plan is available [here](#).

The analytical process began with a descriptive analysis to identify trends, seasonal variations, and irregular patterns using historical quarterly prison population data between 2010 and 2024.

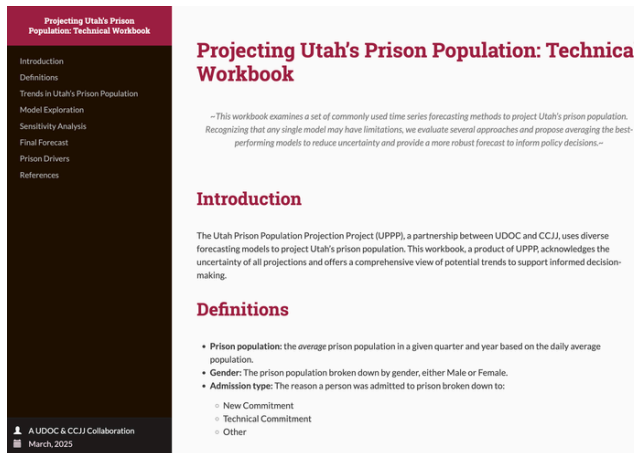
For the *2025 annual update*, prison projections used quarterly UDC daily average population data from 2010 Q1 through 2026 Q2. **The UPPP Dashboard and Annual Report** forecasting model used a two-step approach that accounts for major shifts in Utah's prison population, including those following the Justice Reinvestment Initiative (JRI) and COVID-19 pandemic. Weighted model forecasts were used to develop projection scenarios. Utah's general population growth and violent crime trends were also reviewed as potential influences on future growth. The methodology is formalized in the UPPP Technical Workbook and visualized in Figure 1.

Process & Data

Utah developed its prison population forecasting framework through a comprehensive review of the national literature and consultations with local expertise and selected states to understand their successes and challenges in prison projection work. This review guided the exploration of commonly used forecasting methods to be analyzed on Utah data.



The UPPP methodology was selected through a review of the literature, local expertise and learning from other states.

Figure 1: The UPPP Technical Workbook

**Ongoing validation
improves accuracy for
effective policy decisions.**

refinement will support projection accuracy and informed policy decisions. Future updates may further improve forecasting precision through simulation-based approaches.

Training & Validation

Model performance was evaluated by comparing 2025 UPPP projections with actual prison population outcomes. Forecasts generally performed best in the short term, with the male population beginning to exceed projections in the middle to latter part of 2025. These results informed adjustments to better reflect recent population trends and model performance. The final forecast was then derived by summing the individual gender-based forecasts.

Limitations & Updates

Current projections do not account for significant future legislative or policy changes. The Technical Workbook provides a framework for updating projections as new data emerges. Ongoing validation and

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